SPECTRAL CORRESPONDENCE FOR CYCLIC HIGGS BUNDLES

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Abstract. In this paper, we describe the spectral correspondence for cyclic Higgs bundles from the viewpoint of quiver bundles. Under this framework, we establish a one-to-one correspondence between cyclic Higgs bundles on a curve and sheaves on a noncommutative surface whose noncommutative structure orginates from the path algebra associated to the cyclic quiver. As applications, this correspondence generalizes the known spectral correspondence for $U(p,p)$-Higgs bundles and establish a connection between the spectral data for $U(p,q)$-Higgs bundles and modules over the sheaf of even Clifford algebras of a conic fibration.

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1. Introduction

A cyclic Higgs bundle on a smooth projective curve C is a $GL(r,\mathbb{C})$-Higgs bundle consisting of a vector bundle $E = \bigoplus_{i\in\mathbb{Z}/m\mathbb{Z}} E_i$ and a Higgs field $\Phi : E \to E \otimes K_C$ of the form

$$\Phi = \begin{pmatrix} 0 & 0 & \cdots & 0 & \phi_{m-1} \\ \phi_0 & 0 & & & 0 \\ 0 & \phi_1 & & & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & \cdots & & \phi_{m-2} & 0 \end{pmatrix} \tag{1}$$

where $\phi_i : E_i \to E_{i+1} \otimes K_C$ for $i \in \mathbb{Z}/m\mathbb{Z}$.

This notion arose originally in the work of Simpson [Sim09] (called cyclotomic Higgs bundles) and was defined as the fixed points for the action of $\mathbb{Z}/m\mathbb{Z}$ on the Higgs field. From the perspective of moduli spaces, the moduli space of cyclic Higgs bundles can be viewed as the fixed point locus of an $\mathbb{Z}/m\mathbb{Z}$ action on the moduli space of $GL(r,\mathbb{C})$-Higgs bundle. Cyclic Higgs bundles have also been investigated in other contexts. When $m = 2$, they are Higgs bundles for real Lie group $U(p,q)$ and the topology of their moduli spaces have been studied extensively using Morse theory by Bradlow, García-Prada and Gothen [BGG03]. Analytically, the work of Baraglia [Bar15] establishes a correspondence between cyclic Higgs bundles and solutions of affine Toda equation. This is later generalized to non-compact surface by

Li–Mochizuki [LM25]. More recently, the concept of cyclic Higgs bundles has been generalized to arbitrary reductive groups through the Lie-theorectic perspective of Vinberg pairs by García-Prada and Gonzáles [Gar24; GG26].

In this paper, we are interested in studying the spectral correspondence for cyclic Higgs bundles. The classical spectral correspondence for $GL(r,\mathbb{C})$-Higgs bundles is originally established by Hitchin [Hit87], and subsequently generalized by Beauville–Narasimhan–Ramanan (BNR) [BNR89] and Simpson [Sim94b]. Roughly speaking, the original viewpoint of Hitchin converts a Higgs bundle into a line bundle on its corresponding spectral curve (when smooth) by taking a pointwise eigenspace decomposition. On the other hand, the viewpoint of BNR treats Higgs bundles as modules over the symmetric algebra $S^\bullet K_C^{-1}$ which then lift to torsion-free sheaves on the spectral curve (when integral). Simpson further interprets these modules as coherent sheaves on the surface $X = \underline{\mathrm{Spec}}(S^\bullet K_C^{-1})$. The two approaches of Hitchin and BNR/Simpson parallel an elementary equivalence in linear algebra: A vector space equipped with an endomorphism is equivalent to a module over the polynomial ring $\mathbb{C}[t]$. This can be naturally generalized to an equivalence in the theory of quiver that we will use: A representation of a quiver is equivalent to a module over its path algebra.

A cyclic Higgs bundle can be seen as a quiver bundle for the cyclic quiver $Q(m)$, or equivalently, a representation of $Q(m)$ in the category of vector bundles.

$$Q(m) = \begin{tikzcd} v_0 \arrow[r, "a_0"] & v_1 \arrow[r, "a_1"] & \dots \arrow[r] & v_{m-1} \arrow[lll, bend right, "a_{m-1}"'] \end{tikzcd}$$

We will adopt the module-theoretic viewpoint which is more flexible when the spectral curves become singular. This means that we first treat a cyclic Higgs bundle as a module over the sheaf of the associated path algebra $A(m)$ (generalizing the symmetric algebra $S^\bullet K_C^{-1}$) on C. This path algebra $A(m)$ is always noncommutative and infinite rank. By performing a central reduction of $A(m)$, we obtain a sheaf of noncommutative algebras $\widehat{A}(m)$ of finite rank on the surface $Y = \underline{\mathrm{Spec}}(S^\bullet K_C^{-m})$. Then we can lift the $A(m)$-modules on C to $\widehat{A}(m)$-modules on Y. By rewriting the $\widehat{A}(m)$-modules in terms of quiver sheaves, we arrive at the following version of spectral correspondece for cyclic Higgs bundles.

Theorem 1.1 (Corollary 3.11)**.** *Fix a dimension vector $\boldsymbol{p} = (p_0, \dots, p_{m-1})$ and a degree vector $\boldsymbol{d} = (d_0, \dots, d_{m-1})$. Let $(\widetilde{C}_0, \dots, \widetilde{C}_{m-1}) \in B(\boldsymbol{p})$ be a fixed collection of spectral curves in Y. The following two categories are equivalent:*

(1) *(Higgs data) The category of cyclic Higgs bundles $(\{E_i\}_{i\in\mathbb{Z}/m\mathbb{Z}}, \{\phi_i\}_{i\in\mathbb{Z}/m\mathbb{Z}})$ on C with dimension vector $\boldsymbol{p}$ and degree vector $\boldsymbol{d}$, such that the spectral curve of $\Phi_i := \phi_{i-1} \circ \cdots \circ \phi_{i+1} \circ \phi_i$ is $\widetilde{C}_i$ for all $i \in \mathbb{Z}/m\mathbb{Z}$.*

(2) *(Spectral data) The category of M_Y-twisted $Q(m)$-quiver sheaves $(\{F_i\}_{i\in\mathbb{Z}/m\mathbb{Z}}, \{\psi_i\}_{i\in\mathbb{Z}/m\mathbb{Z}})$ on Y satisfying:*
   (a) *For each $i \in \mathbb{Z}/m\mathbb{Z}$, F_i is a pure dimension one sheaf on Y that is scheme-theoretically supported on $\widetilde{C}_i$ (i.e. an $\mathcal{O}_{\widetilde{C}_i}$-module) with Euler characteristic $\chi(F_i) = d_i + p_i(1-g)$.*
   (b) *The relation $\psi_{i-1} \circ \cdots \circ \psi_i = \mathrm{Id}_{F_i} \otimes_{\mathcal{O}_Y} \lambda$ holds in $\mathrm{Hom}_{\mathcal{O}_Y}(F_i \otimes_{\mathcal{O}_Y} \pi^* K_C^{-m}, F_i)$ for all $i \in \mathbb{Z}/m\mathbb{Z}$.*

Remark 1.2. As we will establish in Proposition 3.9, the spectral data above is equivalent to an $\widehat{A}(m)$-modules. Therefore, Theorem 1.1 can be stated as a one-to-one correspondence between cyclic Higgs bundles on C and coherent sheaves on the noncommutative surface $(Y, \widehat{A}(m))$.

This equivalence can be seen as a natural generalization of the works of BNR/Simpson in the case of $GL(r, \mathbb{C})$, Schaposink [Sch15] in the case of $U(p, p)$ (which parallels the approach of Hitchin) and Peón-Nieto [Peó15] in the case of $SU(p+1, p)$. There is also a more general approach to abelianization of Higgs bundles for arbitrary quasi-split real groups via cameral cover by the work of Peón-Nieto and García-Prada [GP23]. In Section 4, we study some cases where a further reduction of the spectral data is posssible.

A further advantage of working on the noncommutative surface $(Y, \widehat{A}(m))$ is that it reveals a surprising relation to another noncommutative surface studied in the context of the derived category of a conic fibration. More precisely, we show that the sheaf of algebras $\widehat{A}(m)$ is isomorphic to the sheaf of even Clifford algebras associated to a conic fibration over Y (Proposition 4.1). The noncommutative algebraic varieties with the sheaf of even Clifford algebras $\mathcal{C}\ell_0$ as structure sheaf appear originally in the work of Kuznetsov on homological projective duality [Kuz08]. For example, for a vector space W, one of the main results in loc. cit. is that the homological projective dual of $\mathbb{P}(W)$ in the double Veronese embedding $\mathbb{P}(W) \to \mathbb{P}(S^2W)$ is the noncommutative algebraic variety $(\mathbb{P}(S^2W), \mathcal{C}\ell_0)$.

The idea of encoding spectral data in terms of sheaves on surfaces with extra structure has also been explored for other variants of Higgs bundles. Notably, the use of noncommutative geometry and Koszul duality on surfaces to establish spectral correspondence was developed by Katzarkov, Orlov and Pantev for twisted and curved Higgs bundles [KOP]. On the other hand, in the context of parabolic Higgs bundles, one can establish a spectral correspondence in terms of blow-up of ruled surface (see [LL25]). Further blow-up patterns are required if one is interested in parabolic Higgs bundles with irregular singularities [KS14a; Sza17; DDP18]. Alternatively, one can encode the parabolic structure in terms of root stacks and orbifold surfaces [Chu+15].

1.1. **Conventions.** Throughout this paper, we work over the field of complex numbers $\mathbb{C}$. Moreover, all modules over an algebra are assumed to be right modules unless explicitly stated otherwise.

2. Quiver sheaves and modules over the sheaf of path algebras

2.1. **Quiver Sheaves.** In this section, we recall the framework of quiver sheaves and modules over the associated sheaf of path algebras, following [ÁG03]. While we focus entirely in the case of curves, this framework is naturally defined over any algebraic variety.

Let $Q = (Q_0, Q_1)$ be a fixed quiver where Q_0 and Q_1 are the sets of vertices and arrows together with the head and tail maps:

$$h, t : Q_1 \to Q_0.$$

Let C be a smooth projective curve and $M = \{M_a\}_{a \in Q_1}$ be a collection of line bundles on C. An M-twisted Q-quiver sheaf is a pair $(\mathcal{E}, \Phi)$ consisting of a collection of coherent $\mathcal{O}_C$-modules $\mathcal{E} = \{E_i\}_{i \in Q_0}$ and a collection of morphisms $\Phi = \{\phi_a\}_{a \in Q_1}$ where

$$\phi_a : E_{ta} \otimes M_a \to E_{ha}.$$

When E_i are all locally free, we will call it an M-twisted Q-quiver bundle and denote by $\boldsymbol{v} = (\dim(E_i))_{i \in Q_1}$ the dimension vector of the quiver bundle.

Examples 2.1. A $GL(r,\mathbb{C})$-Higgs bundle $(E, \phi : E \to E \otimes K_C)$ on C is an M-twisted Q-quiver bundle with dimension vector $\boldsymbol{v} = (r)$, $M = \{K_C^{-1}\}$, and

$$Q = \begin{tikzcd} v \arrow[loop above] \end{tikzcd}$$

Examples 2.2. A $U(p,q)$-Higgs bundle on C is an M-twisted Q-quiver bundle with dimension vector $\boldsymbol{v} = (p,q)$, $M = \{M_{a_1} = K_C^{-1}, M_{a_2} = K_C^{-1}\}$ and

$$Q = \begin{tikzcd} v_1 \arrow[r, bend right, "a_1"'] & v_2 \arrow[l, bend right, "a_2"'] \end{tikzcd}$$

Examples 2.3. A holomorphic chain of length k is an M-twisted Q-quiver bundle where $M_{a_i} = K_C^{-1}$ for all arrows $a_i \in Q_1$ and

$$Q = \begin{tikzcd} v_1 \arrow[r, "a_1"] & v_2 \arrow[r, "a_2"] & \dots \arrow[r, "a_{k-1}"] & v_k \end{tikzcd}$$

They appear as the fixed points of the $\mathbb{C}^*$-action on the moduli space of $GL(r,\mathbb{C})$-Higgs bundles (corresponding to complex variations of Hodge structure).

2.2. **Path algebras.** Recall that for each quiver $Q = (Q_0, Q_1)$, a path (of length m) is a sequence $a_1 \cdots a_k$ of arrows $a_i \in Q_1$ such that $t(a_i) = h(a_{i+1})$ for $i = 1, \dots, k-1$. For each vertex $i \in Q_0$, we define the trivial path e_i (of length 0).

Let $S = \bigoplus_{i \in Q_0} \mathcal{O}_C \cdot e_i$ be a commutative $\mathcal{O}_C$-algebra generated by the formal symbols e_i with relations

$$\begin{cases} e_i \cdot e_{i'} = e_i, & \text{if } i = i' \\ e_i \cdot e_{i'} = 0, & \text{otherwise.} \end{cases}$$

for each $i, i' \in Q_0$. Moreover, let $\widetilde{M} = \bigoplus_{a \in Q_1} M_a$ and define a S-bimodule structure on it as follows: the left module structure is given by

$$\begin{cases} e_i \cdot m = m, & \text{if } m \in M_a \text{ and } i = ha \\ e_i \cdot m = 0, & \text{otherwise} \end{cases}$$

for all $i \in Q_0, a \in Q_1, m \in M_a$, and likewise for the right module structure. Following [ÁG03], we define the sheaf of M-twisted path algebra A of Q as the tensor S-algebra of the S-bimodule $\widetilde{M}$:

$$A = \bigoplus_{\ell \geq 0} \widetilde{M}^{\otimes_S \ell}$$

where we view $\widetilde{M}^{\otimes 0} := S$.

Equivalently, one can directly define an $\mathcal{O}_C$-algebra structure on

$$MQ := \bigoplus_{\text{all paths } p} M_p$$

where $M_p = M_{a_1} \otimes \dots \otimes M_{a_k}$ and the relation is given by

$$\begin{cases} m_p \cdot m_{p'} = m_p \otimes m_{p'}, & \text{if } tp = hp' \\ m_p \cdot m_{p'} = 0, & \text{otherwise} \end{cases}$$

where $m_p \in M_p, m_{p'} \in M_{p'}$ for all paths p, p'. Then it is straightforward to see that the $A \cong MQ$ as $\mathcal{O}_C$-algebra e.g. the algebra A has relations such as $m_a \otimes m_{a'} = m_a \otimes e_{h(a')} m_{a'} = m_a \cdot$

$e_{h(a')} \otimes m_a$ which vanishes when $t(a) \neq h(a')$. The unit element in A admits a decomposition into orthogonal idempotents

$$1_A = \sum_{i \in Q_0} e_i. \tag{2}$$

Just as the category of quiver representations is equivalent to the category of modules over the associated path algebra, there is an analogue for quiver sheaves. Fix a quiver Q, a coherent right module over the associated sheaf of path algebra A (or simply right A-modules) is an $\mathcal{O}_C$-module E equipped with a right A-module structure given by an $\mathcal{O}_C$-module morphism $E \otimes_{\mathcal{O}_C} A \to E$ satisfying the axioms of right modules over an algebra.

Proposition 2.4. *There is an equivalence between the category of M-twisted Q-quiver bundles on C and the category of coherent right A-modules* mod_C*-A.*

Proof. Given a M-twisted Q-quiver bundles $(\{E_i\}_{i \in Q_0}, \{\phi_a\}_{a \in Q_1})$, we associate to it $E = \bigoplus_{i \in Q_0} E_i$ with a right module structure $E \otimes_{\mathcal{O}_C} A \to E$ induced by the maps ϕ_a which are generators of the path algebra A. This construction defines a functor in the forward direction. Conversely, given a right A-module $\mathcal{E}$ equipped with an A-module structure $\mu_A : \mathcal{E} \otimes_{\mathcal{O}_C} A \to \mathcal{E}$, the decomposition of the unit 1_A induces a decomposition of $\mathcal{E} = \oplus_{i \in Q_0} E_i$. Restricting μ_A to $\mathcal{E} \otimes_{\mathcal{O}_C} \widetilde{M}$ induces the arrow morphisms $\phi_a : M_a \otimes E_{ta} \to E_{ha}$ for $a \in Q_1$. For details, we refer the readers to [ÁG03, Proposition 5.2]. □

3. Spectral correspondence for cyclic Higgs bundles

Definition 3.1. *A cyclic Higgs bundle on C with dimension vector $(p_0, \ldots, p_{m-1})$ is a M-twisted $Q(m)$-quiver bundle with dimension vector $(p_0, \ldots, p_{m-1})$, $M = \{M_{a_i} = K_C^{-1}\}_{i \in \mathbb{Z}/m\mathbb{Z}}$ and*

$$Q(m) = \begin{tikzcd} v_0 \arrow[r, "a_0"] & v_1 \arrow[r, "a_1"] & \ldots \arrow[r] & v_{m-1} \arrow[lll, bend right, "a_{m-1}"'] \end{tikzcd}$$

Remark 3.2. Clearly, a cyclic Higgs bundle in the definition is equivalent to a $GL(\sum p_i, \mathbb{C})$-Higgs bundles whose Higgs field is of the form 1 stated in the introduction.

Let $A(m)$ be the sheaf of M-twisted path algebras on C associated to the quiver $Q(m)$. As $A(m)$ is noncommutative, we begin by studying its central subalgebra $Z(A(m))$ which is commutative. Note that we have the natural direct sum decomposition

$$A(m) = \bigoplus_{i,j \in Q_0} A_{i \leftarrow j} \tag{3}$$

where $A_{i \leftarrow j} := e_i A(m) e_j$ corresponds to paths from v_j to v_i.

Lemma 3.3. $Z(A(m)) \subset \bigoplus_{i \in \mathbb{Z}/m\mathbb{Z}} A_{i \leftarrow i} \subset A(m)$.

Proof. Indeed, if $z \in Z(A(m))$ and we write $z = \sum_{i,j \in Q_0} z_{i \leftarrow j}$ according to the decomposition 3, then the commutativity $e_k \cdot z = z \cdot e_k$ implies that

$$\sum_{j \in Q_0} z_{k \leftarrow j} = \sum_{i \in Q_0} z_{i \leftarrow k}$$

It follows that only $z_{k \leftarrow j} = 0$ for $k \neq j$. □

For each vertex $i \in \mathbb{Z}/m\mathbb{Z}$, we define the path (loop) of length m

$$c_i = a_{i-1} \cdot a_{i-2} \ldots a_{i+1} \cdot a_i$$

where all subscripts are understood modulo m. Clearly, we have

$$A_{i\leftarrow i} = \bigoplus_{\ell=0}^{\infty} M_{c_i^\ell} = \bigoplus_{\ell=0}^{\infty} M_{c_i}^{\otimes \ell}$$

as all paths starting and ending at the vertex v_i must be a composition of the loop c_i. Since $M_{c_i} = K_C^{-m}$, it induces a canonical isomorphism $\iota_i : S^\bullet(K_C^{-m}) \xrightarrow{\sim} A_{i\leftarrow i}$. Let

$$\begin{aligned} \Delta : S^\bullet(K_C^{-m}) &\to \bigoplus_{i\in\mathbb{Z}/m\mathbb{Z}} A_{i\leftarrow i} \subset A(m), \\ s &\mapsto \sum_{i\in\mathbb{Z}/m\mathbb{Z}} \iota_i(s) \end{aligned} \tag{4}$$

be the diagonal map and denote its image by $A_\Delta \subset A(m)$.

Proposition 3.4. *The central subalgebra $Z(A(m))$ is equal to A_Δ. Consequently,*

$$Z(A(m)) \cong S^\bullet(K_C^{-m})$$

as $\mathcal{O}_C$-algebras.

Proof. Let $z = \sum_{i\in\mathbb{Z}/m\mathbb{Z}} z_i \in Z(A(m))$ where $z_i \in A_{i\leftarrow i}$ by Lemma 3.3. It remains to impose that z commutes with all local sections $\alpha_i \in M_{a_i}$. By matching the idempotents (heads and tails), the condition $z \cdot \alpha_i = \alpha_i \cdot z$ implies that

$$z_{i+1} \cdot \alpha_i = \alpha_i \cdot z_i \tag{5}$$

which holds in $A_{i+1\leftarrow i}$. If $z_i \in M_{c_i^\ell}$, then the equality forces $z_{i+1} \in M_{c_{i+1}^\ell}$. As paths, we have $c_{i+1}^\ell \cdot a_i = a_i \cdot c_i^\ell$ which induces a canonical isomorphism $M_{c_{i+1}}^{\otimes\ell} \otimes M_{a_i} \cong M_{a_i} \otimes M_{c_i}^{\otimes\ell}$. Since M_{a_i} is a line bundle (locally isomorphic to $\mathcal{O}_C$), the equality 5 holds for all local sections α_j implies that the local sections z_{i+1} and z_i are identified under the isomorphism $\iota_i \circ \iota_{i+1}^{-1}$. Since this holds for all i, all components z_i are the image of a single section $s \in S^\bullet(K_C^{-m})$, so $Z(A(m)) \subset A_\Delta$.

Conversely, it is straightforward to verify that any diagonal element $s \in A_\Delta$ commutes with the idempotents e_i and the arrow generators of $A(m)$ to establish the reverse direction. □

By Proposition 3.4, $A(m)$ is a $S^\bullet(K_C^{-m})$-module. It is now natural to consider the total space of K_C^m:

$$\pi : Y := \underline{\mathrm{Spec}}(S^\bullet(K_C^{-m})) \to C$$

Let

$$A_Y(m) = \pi^* A(m) = \mathcal{O}_Y \otimes_{\pi^{-1}\mathcal{O}_C} \pi^{-1} A(m)$$

be the usual pullback of a coherent module which is also the sheaf of M_Y-twisted path algebras on Y of the quiver Q with twisting line bundles $M_Y := \{M_{Y,a_i} = \pi^*(K_C^{-1})\}_{i\in\mathbb{Z}/m\mathbb{Z}}$. Consider the two morphisms of sheaves:

- The composition $K_C^{-m} \hookrightarrow S^\bullet K_C^{-m} \xrightarrow{\Delta} A(m)$ is a morphism of $\mathcal{O}_C$-modules. We denote its pullback by $\nu_1 := \pi^*(\Delta|_{K_C^{-m}})$. Locally, for a section $s \in K_C^{-m}$, this acts as:

$$\nu_1(1 \otimes s) = 1 \otimes \sum_{i\in\mathbb{Z}/m\mathbb{Z}} \pi^{-1}(\iota_i(s))$$

- Let $\lambda : \pi^* K_C^{-m} \to \mathcal{O}_Y$ be the morphism induced by the tautological section which lies in $H^0(Y, \pi^* K_C^m)$. We define

$$\nu_2 : \pi^* K_C^{-m} \xrightarrow{\lambda} \mathcal{O}_Y \to A_Y(m), \quad z \mapsto \lambda(z) \otimes 1$$

Then we define the two-sided ideal I in A_Y generated by the difference of ν_1 and ν_2:

$$I := \langle Im(\nu_1 - \nu_2)\rangle = \left\langle 1 \otimes \sum_{i\in\mathbb{Z}/m\mathbb{Z}} \pi^{-1}(\iota_i(s)) - \lambda(1\otimes s)\otimes 1 \middle| s \in K_C^{-m} \right\rangle$$
$$= \left\langle 1 \otimes \pi^{-1}(\iota_i(s)) - \lambda(1\otimes s)\otimes e_{Y,i} \middle| i \in \mathbb{Z}/m\mathbb{Z}, s \in K_C^{-m} \right\rangle$$

where the last equality is obtained by the decomposition of the unit $1_{A_Y(m)} = \sum_{i\in\mathbb{Z}/m\mathbb{Z}} e_{Y,i}$. We define the sheaf of algebras on Y:

$$\widehat{A}(m) := A_Y(m)/I.$$

Geometrically, the ideal I quotients the path algebra by forcing the algebraic action of a full quiver cycle to be identical to multiplication by the tautological coordinate λ on the surface Y. This is precisely the mechanism that flattens the support of the resulting modules onto the spectral curve.

Proposition 3.5.

(1) *As an $\mathcal{O}_Y$-module, $\widehat{A}(m)$ is locally free of finite rank of m^2.*

(2) *Let $Y_0 \subset Y$ denote the zero section of K_C^m over C. For any point $y \in Y \setminus Y_0$, the fiber $\widehat{A}(m)|_y$ is isomorphic to the matrix algebra* $\mathrm{Mat}_{m\times m}(\mathbb{C})$.

Proof. (1) The $\mathcal{O}_Y$-algebra $A_Y(m)$ is generated by sections corresponding to all paths. Any path p starting from v_j and ending at v_i can be factored as $p = c_i^\ell \cdot p_{ij}$, where p_{ij} is the unique path of length $< m$ from v_j to v_i. Taking the quotient of $A_Y(m)$ by I means that we are identifying the action of the loop c_i with multiplication by the tautological section λ. Therefore, in the quotient $\widehat{A}(m) = A_Y(m)/I$, the loop c_i^ℓ just acts as scalar multiplication. It follows that $\widehat{A}(m)$ is generated as an $\mathcal{O}_Y$-module by the sections corresponding to paths of length $< m$. Since the number of paths p of length $< m$ is m^2 and all the component M_p in $A_Y(m)$ is locally free of rank 1, the quotient $A_Y(m)/I$ is locally free of rank m^2.

(2) Let $y \in Y \setminus Y_0$. By the Wedderburn-Artin theorem, it suffices to check that $\widehat{A}(m)|_y$ is a simple $\mathbb{C}$-algebra. Let J be a non-zero two-sided ideal of $\widehat{A}(m)|_y$. For simplicity, we will denote by e_i the restriction of the section e_i of $\widehat{A}(m)|_y$ as well. Take any non-zero element $x \in J$. Write $x = \sum_{i,j\in Q_0} x_{i\leftarrow j}$ according to the direct sum decomposition 3 (which holds for the quotient $\widehat{A}(m)$). Since $x \neq 0$, there must exist a nonzero component $x_{i\leftarrow j} = e_i x e_j$ which also lies in J because J is a two-sided ideal.

Take a nonzero element $u_{j\leftarrow i} \in \widehat{A}(m)|_y$ corresponding to the path from v_i to v_j (which exists because the component of $\widehat{A}(m)|_y$ for any path of length $< m$ is a 1-dimensional vector space). The product $u_{j\leftarrow i} \cdot x_{i\leftarrow j}$ lies in $e_j\widehat{A}(m)|_y e_j \subset J$ which correpsonds to the loop starting and ending at v_j. By the definition of the ideal I of relations, the action corresponding to a loop evaluates to the tautological section $\lambda(y)$ which is non-zero because $y \notin Y_0$. Thus, $u_{j\leftarrow i} \cdot x_{i\leftarrow j} = b_j e_j$ for some non-zero scalar b_j and it follows that $e_j \in J$.

Once we get $e_j \in J$, we can show $e_k \in J$ for all vertex v_k. By choosing a nonzero element $u_{k\leftarrow j}$ and $u_{j\leftarrow k}$ corresponding to the path from v_j to v_k and v_k to v_j respectively. The product $u_{k\leftarrow j} \cdot e_j \cdot u_{j\leftarrow k} \in J$ equals to $b_k e_k$ for some non-zero b_k, which implies that $e_k \in J$. Hence, we obtain all the idempotents e_k and the identity element $1 = \sum_{k\in\mathbb{Z}/m\mathbb{Z}} e_k$. It follows that $J = \widehat{A}(m)|_y$.

□

Remark 3.6. We can view the pair $(Y, \widehat{A}(m))$ as a (midly) noncommutative surface in the sense that the category Mod_Y-$\widehat{A}(m)$ of quasicoherent right $\widehat{A}(m)$-modules on Y is interpreted

as the category $\mathrm{QCoh}(Y, \widehat{A}(m))$ of coherent sheaves on $(Y, \widehat{A}(m))$. Note that while the original sheaf of path algebras $A(m)$ is not a coherent $\mathcal{O}_C$-module (being of infinite rank), its central reduction $\widehat{A}(m)$ on Y has finite rank by Proposition 3.5. Consequently, the category $\mathrm{QCoh}(Y, \widehat{A}(m))$ and its derived category are more well-behaved. We refer the readers to [Kuz08, Section 2][Kuz06, Appendix D] for a detailed account of noncommutative algebraic geometry from this point of view.

Proposition 3.7. *We have an isomorphism of $\mathcal{O}_C$-algebras:*

$$\pi_* \widehat{A}(m) \cong A(m)$$

Proof. Since π is affine, we have $\pi_* \widehat{A}(m) \cong \pi_* A_Y(m)/\pi_* I$. By the projection formula, we have a natural isomorphism:

$$\pi_* A_Y(m) \cong \pi_* \mathcal{O}_Y \otimes_{\mathcal{O}_C} A(m)$$

Recall that $Y = \underline{\mathrm{Spec}}(S^\bullet(K_C^{-m}))$. This implies that $\pi_* \mathcal{O}_Y = S^\bullet K_C^{-m}$ which is isomorphic to the central subalgebra $Z(A(m))$ by Proposition 3.4.

Let $m_C : \pi_* \mathcal{O}_Y \otimes_{\mathcal{O}_C} A(m) \to A(m)$ be the natural multiplication map. Since $\pi_* \mathcal{O}_Y$ maps into the center of $A(m)$, this multiplication is compatible with the algebra structure and m_C is a well-defined homomorphism of $\mathcal{O}_C$-algebras. Furthermore, m_C is clearly surjective since $\pi_* \mathcal{O}_Y$ contains the identity element. It remains to show that the kernel of m_C is the ideal $\pi_* I$.

We can verify this locally over an affine open subset $U = \mathrm{Spec}(R) \subset C$ over which the line bundle K_C^{-m} is trivial. Let $A_R := A(m)(U)$. Over U, the central subalgebra $Z(A(m)) \cong S^\bullet K_C^{-m}$ restricts to $Z_R \cong R[t]$ where t corresponds to a trivializing section of $K_C^{-m}(U)$. The restriction of $\pi_* A_Y(m)$ to U is simply the R-algebra $R[t] \otimes_R A_R$. Locally, the multiplication map m_C becomes

$$m_R : R[t] \otimes_R A_R \to A_R, \quad t \otimes 1 \mapsto \Delta(t)$$

where Δ is the diagonal map defined in 4.

By definition, the ideal $\pi_* I$ is generated by the difference of the maps ν_1 and ν_2. Thus, the ideal $\pi_* I(U)$ is generated by the element $t \otimes 1 - 1 \otimes \Delta(t)$. By standard properties of tensor products of algebras, the kernel of the m_R is generated exactly by the two-sided ideal $\langle t \otimes 1 - 1 \otimes \Delta(t) \rangle$. Therefore, $\ker(m_R) = \pi_* I(U)$. Since this holds for all affine open subset, we conclude that $\ker(m_C) = \pi_* I$.

□

Corollary 3.8. There is an equivalence between $\mathrm{Mod}_Y\text{-}\widehat{A}(m)$ and $\mathrm{Mod}_C\text{-}A(m)$.

Proof. Since $\pi : Y \to C$ is an affine morphism, the pushforward functor $\pi_* : \mathrm{Mod}_Y\text{-}\widehat{A}(m) \to \mathrm{Mod}_C\text{-}\pi_* \widehat{A}(m)$ is an equivalence of categories, just as in the commutative analogue [Sta25, Tag 01SB]. Then the result follows by Proposition 3.7. □

Proposition 3.9. *There is an equivalence between the category $\mathrm{mod}_Y\text{-}\widehat{A}(m)$ of coherent right $\widehat{A}(m)$-modules on Y and the category of M_Y-twisted $Q(m)$-quiver sheaves $(\{F_i\}_{i \in \mathbb{Z}/m\mathbb{Z}}, \{\psi_i\}_{i \in \mathbb{Z}/m\mathbb{Z}})$ with the additional condition that for each $i \in \mathbb{Z}/m\mathbb{Z}$, the composition of arrows around the loop (suppressing the twist by the identity on $\pi^* K_C^{-1}$) yields:*

$$\psi_{i-1} \circ \cdots \circ \psi_i = \mathrm{Id}_{F_i} \otimes_{\mathcal{O}_Y} \lambda \in Hom_{\mathcal{O}_Y}(F_i \otimes_{\mathcal{O}_Y} \pi^* K_C^{-m}, F_i)$$

where the subcript of ψ_i is understood modulo m.

Proof. The category $\mathrm{mod}_Y\text{-}\widehat{A}(m)$ is a full subcategory of $\mathrm{mod}_Y\text{-}A_Y(m)$ consisting of $A_Y(m)$-modules that annihilates the ideal I. By Proposition 2.4, an object $\mathcal{F} \in \mathrm{mod}_Y\text{-}A_Y(m)$ corresponds to a M_Y-twisted $Q(m)$-quiver sheaf $(\{F_i\}_{i\in\mathbb{Z}/m\mathbb{Z}}, \{\psi_i\}_{i\in\mathbb{Z}/m\mathbb{Z}})$. It remains to determine what condition is imposed on this quiver sheaf by requiring it to annihilate I.

Recall that $I = Im(\nu_1 - \nu_2)$. The map $\nu_1 : \pi^* K_C^{-m} \to A_Y(m)$ corresponding to loops in the path algebra induces

$$\mathcal{F} \otimes_{\mathcal{O}_Y} \pi^* K_C^{-m} \to \mathcal{F} \otimes_{\mathcal{O}_Y} A_Y(m) \to \mathcal{F}$$

By restricting this map to the component F_i, we get

$$\psi_{i-1} \circ \cdots \circ \psi_i : F_i \otimes \pi^* K_C^{-m} \to F_i$$

On the other hand, the map $\nu_2 : \pi^* K_C^{-m} \to A_Y(m)$ is defined by multiplication by the tautological section $\lambda \in H^0(Y, \pi^* K_C^m)$. Its induced action on each component F_i maps a section $f \otimes z \in F_i \otimes_{\mathcal{O}_Y} \pi^* K_C^{-m}$ to $f \cdot \lambda(z) \in F_i$, that is,

$$\mathrm{Id}_{F_i} \otimes_{\mathcal{O}_Y} \lambda : F_i \otimes \pi^* K_C^{-m} \to F_i.$$

Therefore, in order for $\mathcal{F}$ to annihilates the ideal I, each component F_i must satisfy

$$\psi_{i-1} \circ \cdots \circ \psi_i = \mathrm{Id}_{F_i} \otimes_{\mathcal{O}_Y} \lambda, \quad \text{for } i \in \mathbb{Z}/m\mathbb{Z}$$

as desired. □

Theorem 3.10. *The following four categories are equivalent:*

(1) *The category of M-twisted $Q(m)$-quiver sheaves $(\{E_i\}_{i\in\mathbb{Z}/m\mathbb{Z}}, \{\phi_i\}_{i\in\mathbb{Z}/m\mathbb{Z}})$ on C.*
(2) *The category of coherent right $A(m)$-modules on C.*
(3) *The category of coherent right $\widehat{A}(m)$-modules on Y whose support is finite over C.*
(4) *The category of M_Y-twisted $Q(m)$-quiver sheaves $(\{F_i\}_{i\in\mathbb{Z}/m\mathbb{Z}}, \{\psi_i\}_{i\in\mathbb{Z}/m\mathbb{Z}})$ on Y such that:*
   (a) *The sheaf F_i has support finite over C for all $i \in \mathbb{Z}/m\mathbb{Z}$.*
   (b) *The relation $\psi_{i-1} \circ \cdots \circ \psi_i = \mathrm{Id}_{F_i} \otimes_{\mathcal{O}_Y} \lambda$ holds in $\mathrm{Hom}_{\mathcal{O}_Y}(F_i \otimes_{\mathcal{O}_Y} \pi^* K_C^{-m}, F_i)$ for all $i \in \mathbb{Z}/m\mathbb{Z}$.*

Proof. The equivalence (1) ⇔ (2) is Proposition 2.4. The equivalence (2)⇔ (3) follows from the Corollary 3.8: under the equivalence π_*, the $\mathcal{O}_C$-coherence condition for an $A(m)$-module translates precisely to the finite support condition for an $\widehat{A}(m)$-module on Y. The equivalence (3) ⇔ (4) is a restriction of Proposition 3.9 to $\mathcal{O}_Y$-coherent sheaves with support finite over C. □

Let $(\mathcal{E}, \Phi) = (\{E_i\}_{i\in\mathbb{Z}/m\mathbb{Z}}, \{\phi_i\}_{i\in\mathbb{Z}/m\mathbb{Z}})$ be a cyclic Higgs bundle. For each $i \in \mathbb{Z}/m\mathbb{Z}$, the composite map for the loop at v_i,

$$\Phi_i := \phi_{i-1} \circ \cdots \circ \phi_{i+1} \circ \phi_i : E_i \otimes K_C^{-m} \to E_i$$

is an $\mathcal{O}_C$-linear map which defines a spectral curve $\widetilde{C}_i = \{\det(\lambda Id - \pi^*\Phi) = 0\} \subset Y$ where $\lambda \in H^0(Y, \pi^* K_C^{-m})$ is the tautological section. The restriction $\pi|_{\widetilde{C}_i} : \widetilde{C}_i \to C$ is a p_i-sheeted cover of C, where $p_i = \mathrm{rank}(E_i)$. The space of spectral curves is parameterized by the coefficients of the characteristic polynomials which lie in the vector space

$$B(p_i) = \bigoplus_{\mu=1}^{p_i} H^0(C, K_C^{m\mu})$$

By a slight abuse of notation, we will write $\widetilde{C}_i \in B(p_i)$ to denote the spectral curve $\widetilde{C}_i$ corresponding to that point in the parameter space $B(p_i)$. For a given dimension vector $\boldsymbol{p} = (p_0, \ldots, p_{m-1})$, let $B(\boldsymbol{p}) = B(p_0) \times \cdots \times B(p_{m-1})$.

Corollary 3.11. Fix a dimension vector $\boldsymbol{p} = (p_0, \ldots, p_{m-1})$ and a degree vector $\boldsymbol{d} = (d_0, \ldots, d_{m-1})$. Let $(\widetilde{C}_0, \ldots, \widetilde{C}_{m-1}) \in B(\boldsymbol{p})$ be a fixed collection of spectral curves in Y. The following two categories are equivalent:

(1) (Higgs data) The category of cyclic Higgs bundles $(\{E_i\}_{i\in\mathbb{Z}/m\mathbb{Z}}, \{\phi_i\}_{i\in\mathbb{Z}/m\mathbb{Z}})$ on C with dimension vector $\boldsymbol{p}$ and degree vector $\boldsymbol{d}$, such that the spectral curve of Φ_i is $\widetilde{C}_i$ for all $i \in \mathbb{Z}/m\mathbb{Z}$.

(2) (Spectral data) The category of M_Y-twisted $Q(m)$-quiver sheaves $(\{F_i\}_{i\in\mathbb{Z}/m\mathbb{Z}}, \{\psi_i\}_{i\in\mathbb{Z}/m\mathbb{Z}})$ on Y satisfying:
   (a) For each $i \in \mathbb{Z}/m\mathbb{Z}$, F_i is a pure dimension one sheaf on Y that is scheme-theoretically supported on $\widetilde{C}_i$ (i.e. an $\mathcal{O}_{\widetilde{C}_i}$-module) with Euler characteristic $\chi(F_i) = d_i + p_i(1-g)$.
   (b) The relation $\psi_{i-1} \circ \cdots \circ \psi_i = \mathrm{Id}_{F_i} \otimes_{\mathcal{O}_Y} \lambda$ holds in $\mathrm{Hom}_{\mathcal{O}_Y}(F_i \otimes_{\mathcal{O}_Y} \pi^* K_C^{-m}, F_i)$ for all $i \in \mathbb{Z}/m\mathbb{Z}$.

Proof. The correspondence between the cyclic Higgs bundles $(\{E_i\}_{i\in\mathbb{Z}/m\mathbb{Z}}, \{\phi_i\}_{i\in\mathbb{Z}/m\mathbb{Z}})$ and quiver sheaves $(\{F_i\}_{i\in\mathbb{Z}/m\mathbb{Z}}, \{\psi_i\}_{i\in\mathbb{Z}/m\mathbb{Z}})$ is the equivalence (1) $\Leftrightarrow$ (4) in Theorem 3.10. In particular, under this equivalence, we have $E_i = \pi_* F_i$ and the Higgs field Φ_i is recovered by pushing forward $\psi_{i-1} \circ \cdots \circ \psi_i = \mathrm{Id}_{F_i} \otimes_{\mathcal{O}_Y} \lambda$ to C. For each i, this reduces precisely to the classical spectral correspondence for $GL(p_i, \mathbb{C})$-Higgs bundles ([BNR89; Sim94b]). Thus, the condition that E_i is locally free of rank p_i and degree d_i with spectral curve $\widetilde{C}_i$ translates directly to F_i being pure dimension one with support $\widetilde{C}_i$ and Euler characteristic $d_i + p_i(1-g)$. □

Remark 3.12 (Constraints on Spectral curves). Suppose $(\widetilde{C}_0, \ldots, \widetilde{C}_{m-1}) \in B(\boldsymbol{p})$ is a fixed collection of spectral curves in Y that comes from a cyclic Higgs bundle. Then the equivalence in Corollary 3.11 imposes a constraint on the spectral curves. Indeed, the relations

$$(\psi_{i-1} \circ \cdots \circ \psi_{i+1}) \circ \psi_i = \mathrm{Id}_{F_i} \otimes_{\mathcal{O}_Y} \lambda, \quad \psi_i \circ (\psi_{i-1} \circ \cdots \circ \psi_{i+1}) = \mathrm{Id}_{F_{i+1}} \otimes_{\mathcal{O}_Y} \lambda$$

implies that $\psi_i : F_i \otimes \pi^* K_C^{-1}|_{Y \setminus Y_0} \to F_{i+1}|_{Y\setminus Y_0}$ is an isomorphism since $\mathrm{Id}_{F_i} \otimes_{\mathcal{O}_Y} \lambda$ is an isomorphism on $Y \setminus Y_0$. Moreover, taking the tensor product with the line bundle $\pi^* K_C^{-1}$ does not change the support of F_i, so F_i and F_{i+1} share the same support away from Y_0. In particular, there exists a common curve $\widetilde{C} \subset Y$ such that $(\widetilde{C}_0, \ldots, \widetilde{C}_{m-1}) \in B(\boldsymbol{p})$ is of the form

$$(\widetilde{C} + q_0 Y_0, \ldots, \widetilde{C} + q_{m-1} Y_0), \quad \text{where } q_0, \ldots, q_{m-1} \in \mathbb{Z}_{\geq 0}. \tag{6}$$

4. Examples

4.1. $GL(r, \mathbb{C})$**-Higgs bundles.** For $m = 1$, $Z(A(1)) \cong A(1)$ and $\widehat{A}(1) \cong \mathcal{O}_Y$. In this case the noncommutative surface $(Y, \widehat{A}(1)) = (Y, \mathcal{O}_Y)$ is just the standard commutative ruled surface Y. The correspondence reduces to the classical result of BNR [BNR89] and Simpson [Sim94b].

4.2. $U(p_0, p_1)$**-Higgs bundles and even Clifford algebras.** Consider $m = 2$ and $\boldsymbol{p} = (p_0, p_1)$. In this case, we will show that the noncommutative surface $(Y, \widehat{A}(2))$ can be identified with another noncommutative surface derived from a conic fibration over Y with ramification locus Y_0.

Let us first recall the definition of conic fibration and its associated sheaf of even Clifford algebras. Let S be a smooth variety. Fix a rank 3 vector bundle F on S and a quadratic form on F which is an embedding of a line bundle $q : L \to S^2F^\vee$. Let $f' : \mathbb{P}(F) := \mathrm{Proj}(\bigoplus_{k\geq 0} S^kF^\vee) \to S$ be the projection map. Since the quadratic form q is equivalent to a section

$$s_q \in H^0(S, S^2F^\vee \otimes L^{-1}) = H^0(\mathbb{P}(F), \mathcal{O}_{\mathbb{P}(F)/S}(2) \otimes f'^*L^{-1}),$$

the zero locus $\mathcal{X}$ of s_q defines a conic fibration $f = f'|_{\mathcal{X}} : \mathcal{X} \to S$ whose fibers are plane conics. The discriminant locus of f is the locus where the rank of the quadratic form drops.

It is known by the work of Kuznetsov [Kuz08] that the bounded derived category $D^b(\mathcal{X})$ of the conic fibration $\mathcal{X}$ over S admits a semiorthogonal decomposition

$$D^b(\mathcal{X}) = \langle D^b(S, \mathcal{C}\ell_0(q)), f^*D^b(S)\rangle$$

where the first component is the bounded derived category of right modules over the sheaf of even Clifford algebra $\mathcal{C}\ell_0(q)$. Just as a quadratic form on a vector space defines the Clifford algebra which decomposes into the even and odd parts, the sheaf of algebras $\mathcal{C}\ell_0(q)$ is obtained as a relative version of this construction. It is the sheaf of $\mathcal{O}_S$-algebras whose fiber at any point $x \in S$ is the even part of the Clifford algebra defined by the quadratic form q_x on the fiber F_x (see the original paper of Kuznetsov [Kuz08] for more details).

The following proposition identifies the sheaf of algebras $\widehat{A}(2)$ as the sheaf of even Clifford algebras of a conic fibration over Y.

Proposition 4.1. *There exists a rank 3 vector bundle F over Y and a quadratic form q on F with discriminant locus Y_0 such that the associated sheaf of even Clifford algebras $\mathcal{C}\ell_0(q)$ is isomorphic to $\widehat{A}(2)$.*

Proof. It is possible to explicitly write down the quadratic form and construct an isomorphism between the two sheaves of algebras. Instead, we will apply a useful criterion of Kuznetsov which essentially reduces to checking that $\widehat{A}(2)$ is an even Clifford algebra over each point $y \in Y$. The idea is to show that $\widehat{A}(2)$ is a pointwise Clifford algebra in the sense of Kuznetsov which implies the existence of the desired quadratic form by [Kuz25, Proposition 2.7]. We will focus on checking the required condition of pointwise Clifford algebra and refer the readers to [Kuz25, Definition 2.5] for a precise definition.

By the analysis in Proposition 3.5, $\widehat{A}(2)$ admits a direct sum decomposition according to the length of paths. More specifically, we have $\widehat{A}(2) \cong \mathcal{O}_Y e_0 \oplus \mathcal{O}_Y e_1 \oplus M_{0\leftarrow 1} \oplus M_{1\leftarrow 0}$ as $\mathcal{O}_Y$-modules where $M_{1\leftarrow 0}$ and $M_{0\leftarrow 1}$ are both isomorphic to $\pi^*K_C^{-1}$. We can define an $\mathcal{O}_Y$-linear trace map via:

$$\mathrm{Tr} : \widehat{A}(2) \to \mathcal{O}_Y, \quad \alpha = \alpha_0 e_0 + \alpha_1 e_1 + \alpha_{0\leftarrow 1} + \alpha_{1\leftarrow 0} \mapsto \alpha_0 + \alpha_1$$

By the definition of $\widehat{A}(2)$, the sections corresponding to loops are identified with $\mathcal{O}_Y$-scalar multiplication by the tautological section λ. So, for any $\alpha_{0\leftarrow 1} \in M_{0\leftarrow 1}$ and $\alpha'_{1\leftarrow 0} \in M_{1\leftarrow 0}$, their products yield $\alpha_{0\leftarrow 1}\alpha'_{1\leftarrow 0} = \lambda e_0$ and $\alpha'_{1\leftarrow 0}\alpha_{0\leftarrow 1} = \lambda e_1$. The trace of their commutator

$$\mathrm{Tr}([\alpha_{0\leftarrow 1}, \alpha'_{1\leftarrow 0}]) = \mathrm{Tr}(\lambda e_0 - \lambda e_1) = \lambda - \lambda = 0.$$

This implies that the trace map also vanishes on $[\widehat{A}(2), \widehat{A}(2)]$.

Moreover, Tr is clearly surjective and the section $\frac{1}{2}(e_0 + e_1) : O_Y \to \widehat{A}(2)$ provides a splitting. Hence, we have a direct sum decomposition

$$\widehat{A}(2) = \mathcal{O}_Y(e_0 + e_1) \oplus \widehat{A}(2)^0 \tag{7}$$

where $\widehat{A}(2)^0 = \ker(\mathrm{Tr})$ contains the commutator subsheaf $[\widehat{A}(2), \widehat{A}(2)]$. This verifies the first half of the definition of a pointwise Clifford algebra.

For any point $y \in Y_0$, the fiber $\widehat{A}(2)|_y$ is a path algebra of the quiver $Q(2)$ with relations. It is observed in [Lee23, Proposition 4.1] that $\widehat{A}(2)|_y$ is isomorphic to the even Clifford algebra associated to a degenerate quadratic form. Moreover, the explicit isomorphism provided in [Lee23, Eq. (19)] identifies the direct sum decomposition 7 with the natural decomposition of the even Clifford algebra.

For any point $y \in Y \setminus Y_0$, the fiber $\widehat{A}(2)|_y$ is a matrix algebra by Proposition 3.5(2) which is the even Clifford algebra associated to a nondegenerate quadratic form (e.g. see [Kuz25, Remark 2.4]). Moreover, the trace map on a matrix algebra is unique, and so is the direct sum decomposition 7. Hence, the isomorphism between $\widehat{A}(2)|_y$ and $\mathrm{Mat}_{2\times 2}(\mathbb{C})$ identifies the the decompositions.

Therefore, $\widehat{A}(2)$ is a pointwise Clifford algebra and the result follows from applying [Kuz25, Proposition 2.7]. Moreover, by its construction, the vector bundle F is given by $(\widehat{A}(2)^0)^\vee$. $\square$

From the perspective of even Clifford algebras, there is a further central reduction of the sheaf of algebras $\widehat{A}(2) \cong \mathcal{C}\ell_0(q)$ into a sheaf of Azumaya algebras $\widehat{\mathcal{C}\ell_0(q)}$ on the 2-nd root stack $\widehat{Y}$ of Y along Y_0 [Kuz08, Section 3.6] (see [Lee23, Section 3.1-3.2] for a summary). In particular, we have

$$\mathrm{Coh}(\widehat{Y}, \widehat{\mathcal{C}\ell_0(q)}) \cong \mathrm{Coh}(Y, \mathcal{C}\ell_0(q)) \tag{8}$$

Note that there is a natural morphism $S^\bullet K_C^{-2} \to S^\bullet K_C^{-1}$ of $\mathcal{O}_C$-algebras which induces a (2-sheeted) cyclic cover $\eta : \widetilde{Y} = \underline{\mathrm{Spec}}(S^\bullet K_C^{-1}) \to \underline{\mathrm{Spec}}(S^\bullet K_C^{-2}) = Y$ with a $\mathbb{Z}/2\mathbb{Z}$-action and ramified at Y_0. So, $\widehat{Y}$ is the quotient stack $\left[\widetilde{Y}/(\mathbb{Z}/2\mathbb{Z})\right]$. If we restrict the equivalence 8 to a smooth curve $\widetilde{C} \subset Y$ which intersects Y_0 transversally (which happens for the case in Section 4.3), then the sheaf of Azumaya algebra splits [Kuz08, Corollary 3.16] and the equivalence 8 becomes

$$\mathrm{Coh}\left([\Sigma/(\mathbb{Z}/2\mathbb{Z})]\right) \cong \mathrm{Coh}(\widetilde{C}, \mathcal{C}\ell_0(q)|_{\widetilde{C}})$$

where $\Sigma \subset \tilde{Y}$ is the double cover of $\widetilde{C}$. When $\widetilde{C} \subset Y$ is the spectral curve for the cyclic Higgs bundles of dimension vector (p,p), the double cover $\Sigma \subset \widetilde{Y}$ is the spectral curve of the associated $GL(2p,\mathbb{C})$-Higgs bundle. Hence, the equivalence implies that cyclic Higgs bundles of dimension vector (p,p) with spectral curve $\widetilde{C}$ corresponds to $\mathbb{Z}/2\mathbb{Z}$-equivariant line bundles on Σ. This description is consistent with the characterization of $U(p,p)$-Higgs bundles as fixed points of an involution (see [Sch15, Proposition 3]).

Remark 4.2. On a related note, we remark that conic fibrations also arise in the context of $PGL(r,\mathbb{C})$-Hitchin systems. By the work of Diaconescu–Donagi–Pantev [DDP07] (see also [KS14b, Section 8.8] for the affine conic fibration description), there is a family of (Calabi-Yau) affine conic fibrations whose intermediate Jacobians are isomorphic to the Hitchin fibers of $PGL(r,\mathbb{C})$-Higgs bundles. In this case, the discriminant locus of the affine conic fibration is the corresponding spectral curve.

4.3. **Spectral data for cyclic Higgs bundles of dimension vector** $(p,\dots,p)$**.** Consider $\boldsymbol{p} = (p,\dots,p)$ and fix $(\widetilde{C}_0,\dots,\widetilde{C}_{m-1}) \in B(\boldsymbol{p})$. By Remark 3.12, the degree of $\widetilde{C}_i$ over C is p and so $q_0 = \cdots = q_{m-1} = 0$ in 6. This means that the associated spectral curves $\widetilde{C}_0 = \cdots = \widetilde{C}_{m-1} = \widetilde{C}$ coincide.

Suppose the common spectral curve $\widetilde{C}$ is smooth and not equal to Y_0. The spectral data $(\{F_i\}_{i\in\mathbb{Z}/m\mathbb{Z}}, \{\psi_i\}_{i\in\mathbb{Z}/m\mathbb{Z}})$ in Corollary 3.11 can be further simplified as follows. For $i \in \mathbb{Z}/m\mathbb{Z}$, the pure dimension one sheaf F_i on Y is a line bundle L_i on $\widetilde{C}$. Note that

$$\psi_i \in \mathrm{Hom}_{\widetilde{C}}(L_i \otimes_{\mathcal{O}_{\widetilde{C}}} \pi^* K_C^{-1}, L_{i+1}) \cong H^0(\widetilde{C}, L_i^{-1} \otimes L_{i+1} \otimes \pi^* K_C).$$

Since ψ_i is an isomorphism away from Y_0, we see that ψ_i is a non-zero section of $L_i^{-1} \otimes L_{i+1} \otimes \pi^* K_C$ and determines a nonnegative degree divisor $D_i = \mathrm{div}(\psi_i)$. Conversely, given L_i and a nonnegative degree divisor D_i, we can recover L_{i+1} as $\mathcal{O}(D_i) \otimes L_i \otimes \pi^* K_C^{-1}$ and ψ_i as the unique section that vanishes on D_i. The relation $\psi_{i-1} \circ \cdots \circ \psi_i = \mathrm{Id}_{L_i} \otimes_{\mathcal{O}_Y} \lambda$ translates into the relation that

$$D_0 + \cdots + D_{m-1} = \mathrm{div}(\lambda) \tag{9}$$

By starting with a line bundle L_0 on $\widetilde{C}$ and running the argument above, we conclude that the spectral data $(\{F_i\}_{i \in \mathbb{Z}/m\mathbb{Z}}, \{\psi_a\}_{a \in \mathbb{Z}/m\mathbb{Z}})$ in Corollary 3.11 is equivalent to the data of $(L_0, D_0, \ldots, D_{m-1})$ where L_0 is a line bundle on $\widetilde{C}$ and each D_i is a divisor on $\widetilde{C}$ of nonnegative degree satisfying the relation 9.

For $\boldsymbol{p} = (p, p)$, the description above recovers the spectral data for $U(p,p)$-Higgs bundles studied in the work of Schaposnik [Sch15].

5. Future Directions

5.1. **Spectral data for cyclic Higgs bundles of arbitrary dimension vector $(p_0, \ldots, p_{m-1})$.** The spectral data in the case of $(p, \ldots, p)$ admits a simplification to $(L_0, D_0, \ldots, D_{m-1})$ because the associated spectral curves $\widetilde{C}_0 = \cdots = \widetilde{C}_{m-1}$ coincide. However, as soon as one of the components in the dimension vector is different from the others, at least one of the spectral curves will be reducible.

For example, consider $\boldsymbol{p} = (p+1, \ldots, p)$ and fix $(\widetilde{C}_0, \ldots, \widetilde{C}_{m-1}) \in B(\boldsymbol{p})$. As noted in Remark 3.12, we must have $q_0 = 1$ and $q_i = 0$ for $i \neq 0$ in 6. This implies that the associated spectral curves are given by $\widetilde{C}_0 = \widetilde{C} \cup Y_0$ and $\widetilde{C}_1 = \cdots = \widetilde{C}_{m-1} = \widetilde{C}$, where $\widetilde{C}$ is a degree p cover over C. Then it is possible to reduce the spectral data in Corollary 3.11 in terms of spectral data on $\widetilde{C}$ for dimension vector $(p, \ldots, p)$, a line bundle on $Y_0 \cong C$ and a gluing data (such as [Peó15, Corollary 5.10]). However, the complexity of this kind of analysis grows as the difference between the components in the dimension vector increases.

5.2. **Moduli spaces of cyclic Higgs bundles.** As mentioned in the introduction, the moduli space of cyclic Higgs bundles can be viewed as the fixed point locus of an $\mathbb{Z}/m\mathbb{Z}$ action on the moduli space of $GL(r, \mathbb{C})$-Higgs bundle. By the spectral corresondence established in Corollary 3.11 and Proposition 3.9, we can interpret this as the moduli space of pure dimension one sheaves on the noncommutative surface $(Y, \widehat{A}(m))$. One can argue the existence of such moduli space by defining the appropriate stabiltiy conditions and show that the sheaf $\widehat{A}(m)$ of algebras is a sheaf of rings of differential operators in the sense of Simpson [Sim94a]. Then the general theory of Simpson guarantees the existence of moduli space of semistable $\widehat{A}(m)$-modules on Y.

A potential application of this reinterpretation is an algebraic approach to study their connected components. Recall from Section 4.2 that $\widehat{A}(m)$ is isomorphic to the sheaf of even Clifford algebras of a conic fibration on Y. It is proven in the work of Lahoz–Macrì–Stellari [LMS15, Theorem 2.12] that the moduli space of modules over even Clifford algebras which arises from a cubic threefold is irreducible and hence connected. Their proof is a variant of the classical argument of Mukai that originally establishes the connectedness of the moduli space of sheaves on a K3 surface. Therefore, we expect a variant of Mukai's argument can be used to prove the connectedness of the moduli space of $\widehat{A}(m)$-modules on Y (after imposing a suitable numerical invariant such as the Toledo invariant). This would provide an algebraic approach that complements the Morse-theoretic approach of Bradlow–García-Prada–Gothen [BGG03] to study the connected components of the moduli space of $U(p,q)$-Higgs bundles.

5.3. **Generalized Hitchin fibration for quiver bundles.** From the quiver theoretic viewpoint, cyclic Higgs bundles represent the quiver bundles of a specific quiver $Q(m)$. It would be interesting to study the spectral correspondence for quiver bundles of other types of quivers. A quick observation is that the reason that the case of the quiver $Q(m)$ works well is that the center of the associated path algebra is exactly $S^\bullet K_C^{-m}$ (Proposition 3.4). It is known from the general theory of quiver that a quiver without oriented loop has trivial center. So, the natural first step to analyze the case for other quivers will be looking at quivers with oriented loops and study the center subalgebra of their path algebras.

From the moduli viewpoint, the spectral correspondence for quiver bundles is equivalent to the study of the fibers of the corresponding generalized Hitchin fibration (originally defined by Schmitt in [Sch05]). To give a brief description of the generalized Hitchin fibration, it is more convenient to use the language of stacks.

Let $Q = (Q_0, Q_1)$ be a general quiver with head and tail maps $h, t : Q_1 \to Q_0$. For a fixed dimension vector $\boldsymbol{p} = (p_i)_{i \in Q_0}$, we associate to each vertex $i \in Q_0$ a complex vector space V_i of dimension p_i. The space of representations of Q with dimension vector $\boldsymbol{p}$ is the affine space

$$\mathrm{Rep}(Q, \boldsymbol{p}) = \bigoplus_{a \in Q_1} \mathrm{Hom}(V_{t(a)}, V_{h(a)}).$$

The change of basis group is defined as the product

$$G_{\boldsymbol{p}} := \prod_{i \in Q_0} GL(V_i)$$

which acts naturally on $\mathrm{Rep}(Q, \boldsymbol{p})$ by conjugation. Given this action, there is the associated morphism from the quotient stack to the GIT quotient:

$$\chi : [\mathrm{Rep}(Q, \boldsymbol{p})/G_{\boldsymbol{p}}] \to \mathrm{Rep}(Q, \boldsymbol{p})//G_{\boldsymbol{p}}$$

Just as the original Hitchin fibration for G-Higgs bundles is induced by $[\mathfrak{g}/G] \to \mathfrak{g}//G$, by following the general construction as in [Ngô06; Ngô10], the morphism χ induces a generalized Hitchin fibration from the moduli stack of quiver bundles to an affine Hitchin base.

In fact, Ngô has recently initiated a general framework to study the generalized Hitchin fibration induced from the action of a reductive group on an affine normal scheme (see the related work [HM25]). The action of $G_{\boldsymbol{p}}$ on $\mathrm{Rep}(Q, \boldsymbol{p})$ can be seen as an example of this general framework. His approach is closer to the original cameral cover approach that works well over the regular locus. Our emphasis here is on the quiver structure and the description of the whole Hitchin base. It will be interesting to study the relation between the different approaches.

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